

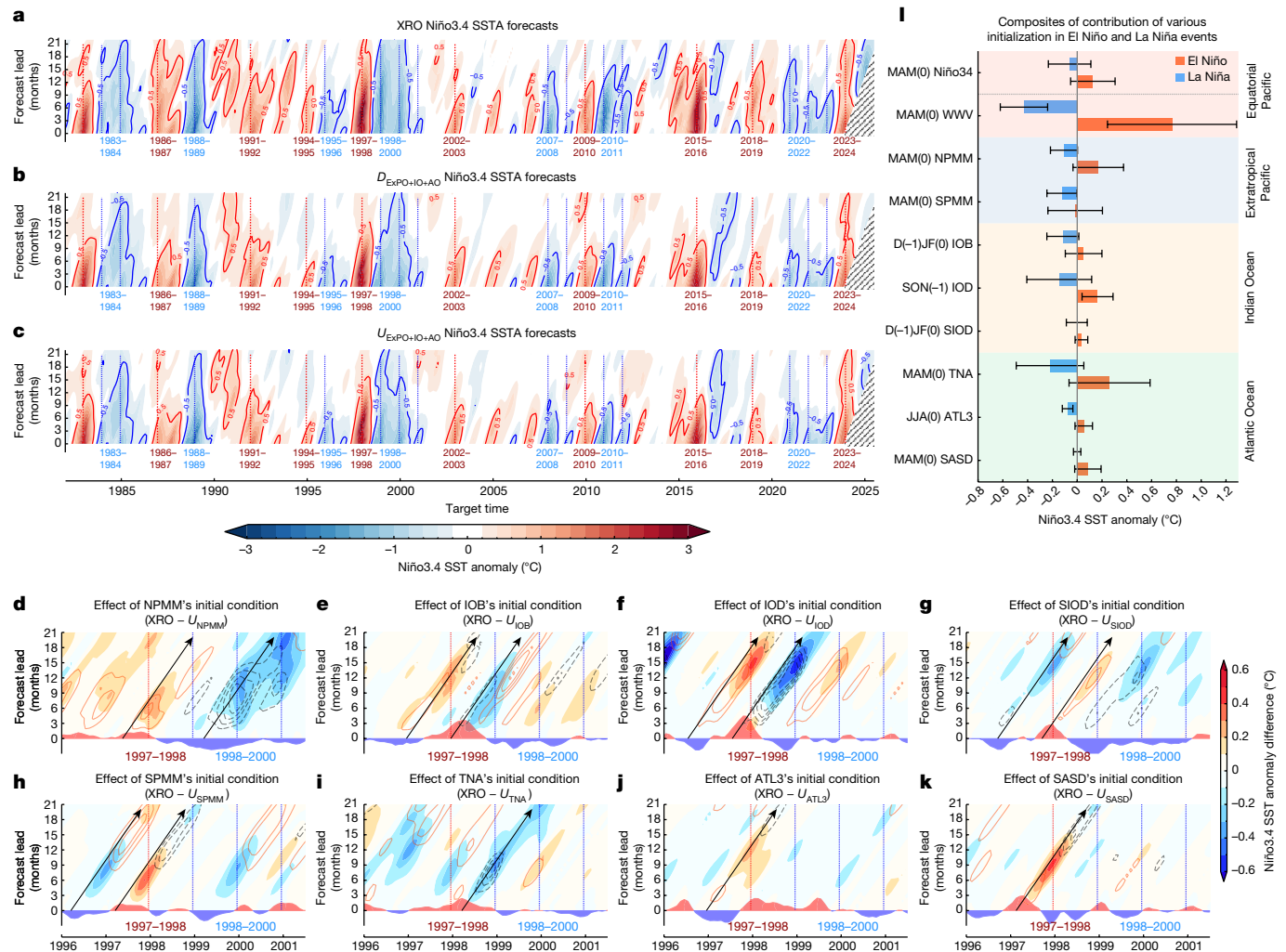


Fig. 4 | Delineating contributions to ENSO amplitudes from other climate modes. **a–c**, Time series of Niño3.4 forecasts for the XRO model (**a**), decoupled ExPO + IO + AO experiment (**b**) and uninitialized ExPO + IO + AO experiment (**c**), as function of target time and forecast lead. **d–k**, The difference of Niño3.4 SSTAs (shading) and WWV anomalies (contours with interval of 0.6 m, positive in red and negative in black dashed, zero omitted), as a function of forecast start month and target month, between the control and uninitialized climate mode experiments for NPMM (**d**), IOB (**e**), IOD (**f**), SIOD (**g**), SPMM (**h**), TNA (**i**), ATL3 (**j**) and SASD (**k**). Vertical reference dashed lines denote December of El Niño (red) and La Niña (blue) years, respectively. The normalized observed time series of each climate mode SSTA index is indicated on the bottom axis; the black arrows indicate the flow of forecast integration started from the

selected time in the bottom. **l**, Composite difference of November to January Niño3.4 SSTA forecasts during El Niño events (red) and La Niña events (blue) between control and uninitialized U_m experiments started from months in a specific preceding season (–1 and 0 in parentheses denote preceding and current year, y axis from top to bottom is $U_{\text{Niño3.4}}$, U_{WWV} , U_{NPMM} , U_{SPMM} , U_{IOB} , U_{IOD} , U_{SIOD} , U_{TNA} , U_{ATL3} and U_{SASD} , respectively); the events are selected when November to January Niño3.4 indices are greater than their standard deviation, which includes seven El Niño events (1982, 1986, 1991, 1997, 2002, 2009, 2015) and five La Niña events (1988, 1998, 1999, 2007, 2010). The error bars show one standard deviation spread among the seven El Niño/five La Niña events. The XRO sensitivity experiments quantify the pathways through which the other climate modes influence El Niño and La Niña events.

ENSO events⁴². For example, the IOB warming in 1998 contributed to a roughly 0.2 °C Niño3.4 SSTA decrease during the 1998–1999 La Niña, about half the magnitude of the IOD-induced change (Fig. 4e). These results corroborate the findings in Fig. 3e that the IO's influence on ENSO predictability is predominantly governed by the IOD.

In the AO, the TNA warming favours a Niño3.4 SSTA decrease 6–12 months later by about 0.3 °C (Fig. 4i), consistent with a previous finding¹⁷. The 1997 boreal summer Atlantic Niña (ATL3 cold anomalies) was found to weakly favour a Niño3.4 SSTA increase 6–12 months later by about 0.15 °C (Fig. 4j). The positive phase of the SASD in 1997 contributed to a roughly 0.3 °C Niño3.4 SSTA increase 9–12 months later (Fig. 4k), in line with previous findings¹⁹. The AO's influence is predominantly governed by the TNA and secondly by the SASD and ATL3.

For the 2020–2022 triple La Niña events, the strong positive IOD in 2019 autumn is among the most important contributors to the first year SSTA cooling (Extended Data Fig. 6d), and the NPMM cooling is

among the most important sources in amplifying the second year SSTA decrease (Extended Data Fig. 6a), consistent with previous findings^{43,44}. The continuing 2023–2024 El Niño occurrence can be predicted up to 18 months in advance in the decoupled ExPO + IO + AO experiment (Fig. 4b), largely due to the highly recharged WWV state caused by the preceding 'triple-dip' La Niña events. The XRO refines the amplitude prediction for the 2023–2024 El Niño at longer lead time of 9–18 months (Fig. 4a), with positive contributions from the preceding IOD and IOB conditions (Extended Data Fig. 6c,d).

Composites of the uninitialized experiments for the peak phase of El Niño and La Niña years (Fig. 4l) support that climate mode interactions contribute to the observed Niño3.4 SSTA anomalies, in addition to the generally stronger contribution from the equatorial Pacific recharge–discharge dynamics intrinsic to ENSO. The extra contributions are mainly from the NPMM, IOD and TNA with large inter-event spread, with other modes playing secondary roles. The impacts are